

Contextual Ontology Extraction and Inspection of Temporal Narrative Networks

Keywords: Natural Language Processing, Knowledge Representation and Reasoning, Large Language Models, Visualization.

Abstract: We present a narrative extraction workflow connecting readers' questions, generated relationships, temporal and attribution details, and visual inspection of evidence and errors. We compare extraction before seeing a question followed by fixed selection with independent contextual extraction on three synthetic stories and three short published passages. Assessment separates strict agreement with reference answers from preserved meaning, relevance and unresolved interpretation. All 12 story responses were parseable. Fixed selection matched every possession and dated location target without extras, whereas contextual answers retained irrelevant facts. On fable actions, strict F1 was 0.625 for selection and 0.308 for contextual extraction. Query-guided extraction of statements and thoughts from an excerpt of Alice's Adventures in Wonderland preserved 3 of 4 semantic targets despite zero strict F1. Prose parsing succeeded in 7 of 9 calls. The workflow makes these mixed results easy to inspect, including useful thought representations, incorrect participants and lost conditional scope. These small development comparisons identify concrete extraction and selection problems for further research.

1 INTRODUCTION

Stories describe people, places and objects whose relationships change over time. Readers may want to know who carries an object, where it is, or what a character believes about it. We present a workflow for exploring narrative ontologies, structured representations of a story's entities and relationships, as temporal networks. Nodes represent characters, objects or places, and edges show their relationships. Temporal details indicate when a relationship holds, while links to the text let readers check its basis. Readers can also examine whether a relationship reflects a character's belief or the narrator's account.

We compare two ways to produce a network for a reader's question. *Pre-extract/select* (A) first extracts relationships without seeing the question, then selects existing records using fixed filters. *Contextual extraction* (B) extracts relationships directly from the same text with the question supplied. Both use records with entity names, evidence citations, time intervals and attribution. Our contributions are:

1. An implemented workflow connecting narrative questions, temporal networks and source evidence. Linked views let readers inspect relationships and trace their meaning to the text (Section 3 and Figures 1-3).
2. An empirical comparison of extraction before and after the question is supplied. Results on synthetic

stories and published prose help distinguish problems in extracting relationships from problems in selecting relevant ones (Section 5).

3. An assessment combining agreement with reference answers, preservation of meaning, relevance and unresolved interpretation. Tables 1 and 2 and the annotated networks show useful partial results alongside errors.

Together, these contributions support the inspection of narrative ontologies beyond a single accuracy score. Small development comparisons show which relationships survive extraction, which details are lost, and how questions affect the resulting networks.

2 RELATED WORK

Extraction and representation. TextRunner extracts relationships without a predefined relation vocabulary and indexes them for later queries (?). SPIRES uses recursive language model prompts to fill a supplied schema and link entities to external vocabularies (?). Building on these approaches, we compare extraction with and without an available question while retaining local entity names and supporting text.

Temporal ontologies provide a foundation for describing change. OWL-Time represents instants, intervals, durations and ordering (?). GRaSP connects

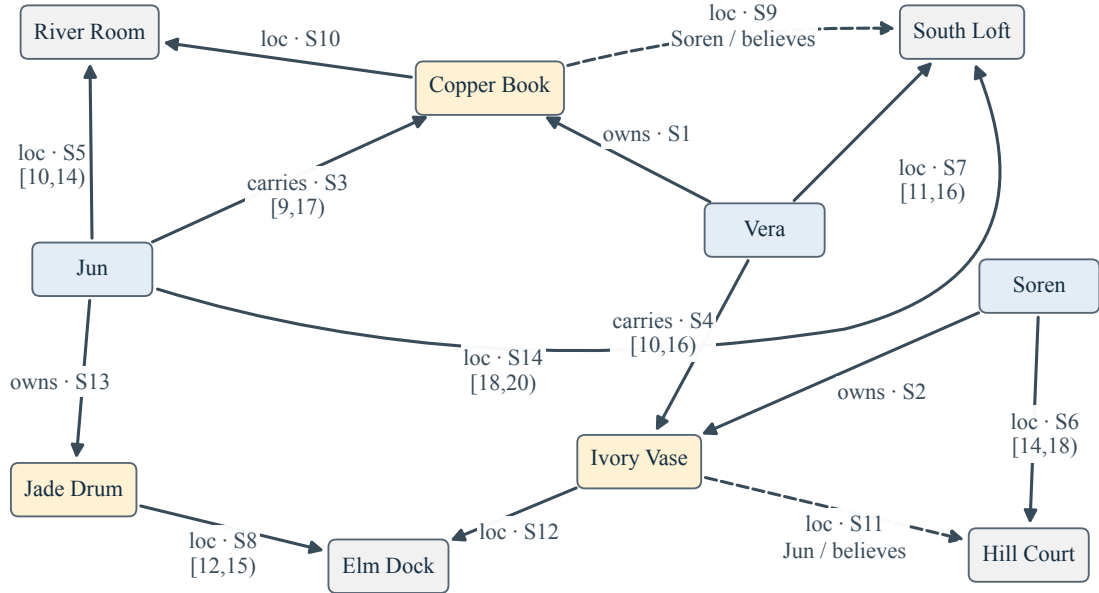
Orchard · text, model network and fixed query selection

S1 Vera owns Copper Book.

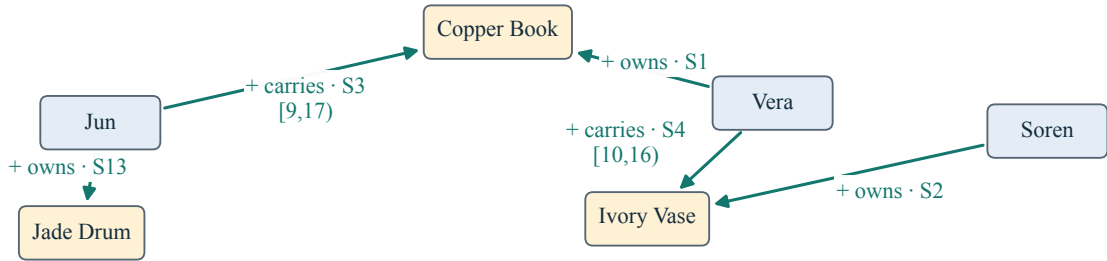
S3 Jun carries Copper Book from day 9 up to but not including day 17.

S11 Jun believes that Ivory Vase is in Hill Court.

Model extraction before the question · 10 nodes / 14 records



Fixed selection: ownership/carrying · 6 nodes / 5 records



Blue fill: person. Cream: object. Grey: place (display categories). loc = located in. Dashed broad edges carry holder / attitude, not extra relationships. + supported. Unprinted bounds and attribution are null. Source excerpted: all 14 sentences supplied.

Figure 1: Orchard: the actual model extraction before the question (top network), followed by fixed selection from that extraction (bottom). Executed question: Who owns or carries what? Return all and only directly narrated ownership and carrying relationships, retaining every stated interval. All 14 extracted records and all five selected records are shown. Source excerpts S1, S3 and S11 come from the complete fourteen-sentence input. Exact endpoint strings share nodes. The selected six-node network is connected. Holder/attitude badges qualify two location edges without adding relationships. The broad extraction is not a gold graph. + denotes retained support for the selected records, not a selection colour. Node categories are display annotations. Horizontal anchors are shared, with compressed vertical spacing below.

statements, textual mentions, sources and attitudes, with an interface linking events to their evidence (?). These foundations inform our compact representation of when relationships hold and who expresses a be-

lief.

Comparison and inspection. RelVis combines extraction comparison, visual inspection and error annotation (?). It also distinguishes valid relationships

missing from reference annotations from extraction errors. CaRB improves reference annotations and scoring through partial coverage and checks on the alignment of relationships and their participants (?). Using our own strict matcher, we report agreement separately from judgments about preserved meaning, missing details and unresolved readings.

Our visualization draws on established filtering, linked selection and coordinated views (?). The contribution brings these ideas together for narrative questions, preserving temporal information and attribution while comparing when extraction occurs relative to the question. Worked examples reveal both useful relationships and errors, such as a correct action linked to the wrong participant. The cited systems provide methodological foundations rather than experimental baselines.

3 REPRESENTATION AND METHODS

3.1 Evidence and Qualified Records

Each request receives the complete selected story or contiguous excerpt with unchanged wording and numbered evidence spans. Each extracted relationship is represented by a qualified record:

$$f = \langle s, r, o, E, t_0, t_1, h, a \rangle. \quad (1)$$

The subject s , relation r and object o define a network edge. Its remaining fields provide evidence identifiers E , temporal bounds t_0 and t_1 , and attribution through a holder h and attitude a . The model supplies every record value. The runtime never fills missing meaning from reference answers.

The following illustration is authored to explain the format and is not a model result:

```
{ "subject": "Nara", "relation": "carries",
  "object": "map", "evidence_ids": ["X1"],
  "valid_from": 2, "valid_until": 5,
  "holder": null, "attitude": null }
```

Given supporting evidence in span X1, this record states that Nara carries the map during [2,5), including the start and excluding the end. A query window [3,4) selects the overlapping record while preserving its original bounds.

Temporal bounds describe intrinsic validity, meaning when the relationship holds. An occurrence or observation alone does not establish its onset or duration. Unsupported bounds remain `null`, indicating unspecified information rather than timeless validity or falsehood. Relative phrases such as “shortly after”

are not converted into numbers, so their temporal detail may remain unrepresented in these bounds. For belief, (s, r, o) describes the believed relationship, h names the believer and $a = \text{believes}$. Direct narration uses `null` for both attribution fields without implying that a character knows the stated relationship. Speech and mental acts may also be explicit relationships, without treating their content as established reality. Pronoun resolution requires a clear referent in the excerpt. The frozen matcher treats missing nullable fields as unspecified, while the field-compliance assessment separately records their absence.

3.2 Extraction Before or With the Question

Pre-extract/select (A) first requests all explicitly supported relationships without showing the questions. Its actual output is frozen before any contextual request is sent. CPU selection retains existing records by relation category, requested entity, attribution status or explicit interval overlap. Synthetic rules distinguish possession, dated location and belief/narration. Prose rules use lexical action and claim categories appropriate to each passage. These fixed rules do not infer relationships, repair meanings or consult reference answers.

Contextual extraction (B) receives the same complete evidence and one question, requesting only relationships needed to answer it. It never consumes A’s graph. Both approaches share the representation, model and authoring conventions. Prompts, questions, selectors and call order are frozen before each batch. All extractions without questions precede contextual transmission. Each request runs once without adaptive repair, independently of another output’s failure.

The experiments use Qwen/Qwen3-8B-AWQ in nonthinking mode (?; ?), served by vLLM (?) on one RTX 4090. Temperature is 0.7, top-p 0.8, top-k 20, and seed 1988649846. Ordinary JSON is requested without a guided schema.

3.3 Inspecting Actual Output

Graph edges retain generated direction, names, citations and qualifications. Only identical endpoint strings share a display identity. Aliases are not inferred, and objects containing sentences remain literal nodes. The Orchard network makes shared participants and connectivity visible (Figure 1). Node fills denote display categories, not model types or correctness. Status symbols and edge styles distinguish supported, partial, unsupported, unresolved and irrel-

evant content. A citation highlight is a navigation aid, not verification. Other figures use relationship rows for long labels. Missing targets appear beside graphs, never as corrected model edges. Display subsets do not change scoring denominators.

4 EVALUATION

4.1 Materials and Annotation

Each evaluation case combines a numbered narrative T_i , a reader’s question q_{ij} and reference records $\mathcal{R}_{ij}$, including accepted alternatives:

$$d_{ij} = \langle T_i, q_{ij}, \mathcal{R}_{ij} \rangle. \quad (2)$$

Evidence, question scope, reference alternatives and normalization rules are frozen before generation, giving both approaches the same evaluation target.

The controlled stories Harbor, Orchard and Museum each contain 14 evidence sentences. Three questions address possession, location within an interval, and belief versus narration. These tasks reveal whether extracted networks preserve relationships together with their temporal and attribution details. Harbor and Orchard informed development. Museum varies sentence order and entity relationships fixed before this batch. Earlier experiments remain separate.

The prose comparison uses Townsend’s complete Lion and Mouse fable (133 words), the opening of *Alice’s Adventures in Wonderland* (112 words), and a contiguous opening passage from *The Red-Headed League* (159 words) (?; ?; ?). Citations identify the retained Gutenberg editions. Passages were selected before execution for identifiable participants, actions, thoughts and dialogue, with two questions per passage addressing actions and claims.

RelVis demonstrates the value of inspecting extraction errors, while CaRB shows how reference annotations and matching choices affect evaluation (?; ?). Building on these findings, our design connects extraction before or after question disclosure to the preservation of relevance, temporal bounds and attribution in narrative networks.

Synthetic inventories distinguish unsupported relationships from supported but irrelevant ones. Prose references cover the requested information. Separate semantic assessments distinguish wording differences from lost or altered meaning. Reference answers and response assessments have not undergone independent human validation. Ambiguous interpretations, including Holmes’s vocative and Alice’s daisy-chain consideration, remain open to author review.

4.2 Agreement, Meaning and Failure

Strict qualified-fact matching requires agreement on the normalized relationship, evidence set, temporal bounds and attribution. Let n be the number of predictions, m the number of reference targets and k the number of one-to-one matches against frozen alternatives. For nonzero denominators,

$$P = \frac{k}{n}, \quad R = \frac{k}{m}, \quad F_1 = \frac{2k}{n+m}. \quad (3)$$

Incorrect, duplicate, irrelevant and unresolved predictions remain in n , while omissions reduce recall. Bare relationship scores ignore citations and qualifications. Strict F1 therefore measures reference agreement, not factual truth.

Semantic assessment separately labels supported, partially supported, unsupported and unresolved content, alongside qualification errors and relevance. Target coverage counts distinct meanings explicitly preserved. One prediction may cover several targets, but partial support is not full coverage. Support requires positive evidence. Missing meaning is never supplied, and equivalence rules remain unchanged after generation.

Strict parsing rejects malformed JSON, duplicate keys and non-JSON constants. A separately recorded recovery may remove only trailing commas immediately before closing delimiters outside strings, preserving raw bytes and an edit record. Neither main batch required recovery. Unparseable outputs have unavailable graph scores and are not scored as intentionally empty graphs.

The analysis is descriptive. These purposively selected development examples share stories and reuse outputs during selection. Records therefore cannot be treated as independent experimental trials, and no significance tests are applied.

5 RESULTS

5.1 Synthetic Selection and Qualifications

All 12 compact-story outputs are strictly parseable. Pre-extract/select (A) matches every possession and dated location target without extras (Table 1). Contextual extraction (B) commonly retrieves the requested physical relationships but returns irrelevant facts in every answer. Orchard’s location answer also omits a target: recall is 0.750. The fresh Museum story shows the same excessive selection as the earlier stories.

The Lion and the Mouse · actions and rescue

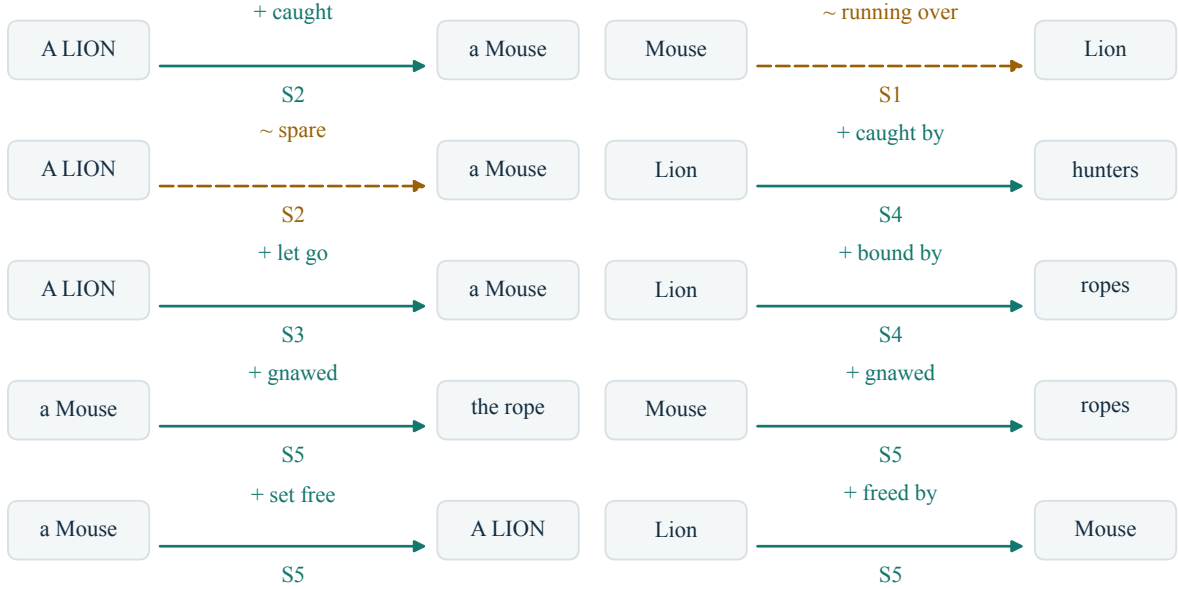
S2 ... "If you would only spare my life, I would be sure to repay your kindness."

S5 The Mouse, recognizing his roar, came and gnawed the rope with his teeth, and set him free, exclaiming:

Request: physical actions, excluding dialogue and intentions

A: text → extraction → selection

B: text + question → extraction



+ supported meaning ~ partial meaning. Citations are not proof of support.

A: spare loses conditional scope. B: early capture/release missing. Source excerpted. Full fable supplied.

Figure 2: Fable actions from pre-extract/select (A) and independent contextual extraction (B). + indicates supported meaning and ~ partial meaning. Display: five of eight A records and all five B records. A's awakening, hunter capture and rope binding are omitted only from display. All predictions remain scored. Executed question: Which physical running/waking, capture, binding, rope-gnawing and release relationships involve the Lion, Mouse, hunters and ropes? Exclude dialogue, intentions, laughter and the moral. Source display excerpts S2 and S5 from the complete supplied fable. The spare edge loses conditional scope and retains the misspelled key `valid until`. B loses the face-specific running endpoint and omits the early capture/release. Other displayed qualifications are null. Assessments are Codex-authored.

Orchard illustrates actual selection from a broader model extraction (Figure 1). Its five ownership/carrying records connect six nodes, retaining explicit carrying intervals. This illustration does not establish a construction effect. The companion contrasts this slice with the twelve-record contextual answer, which also retains seven irrelevant locations.

Among emitted records with identifiable explicitly timed sources, 79 of 79 preserve their full intervals, including irrelevant records. This is conditional preservation, not temporal recall over every reference. The companion's Harbor timeline distinguishes these intervals from the query window. Attribution is less stable: Orchard's pre-extraction decomposes beliefs correctly, whereas Harbor and Museum use sentence-

valued belief objects that disrupt selection.

5.2 Published Prose and Partial Meaning

The prose comparison yields 7 strictly parseable responses from 9 calls. Extra closing braces invalidate Holmes's pre-extraction and contextual action answer. The shared pre-extraction failure blocks both A questions (Table 2). The parseable Holmes claims answer contains partial and unresolved content without fully covered targets.

For fable actions, pre-extract/select performs better on both measures, with strict F1 0.625 versus 0.308 and semantic coverage 6 of 8 versus 4 of 8.

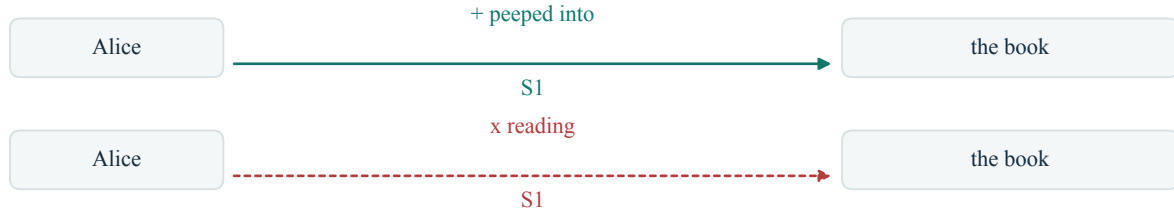
Alice · inspecting an action and a thought

S1 ... once or twice she had peeped into the book her sister was reading ...

S1 ... "and what is the use of a book," thought Alice "without pictures or conversations?" ...

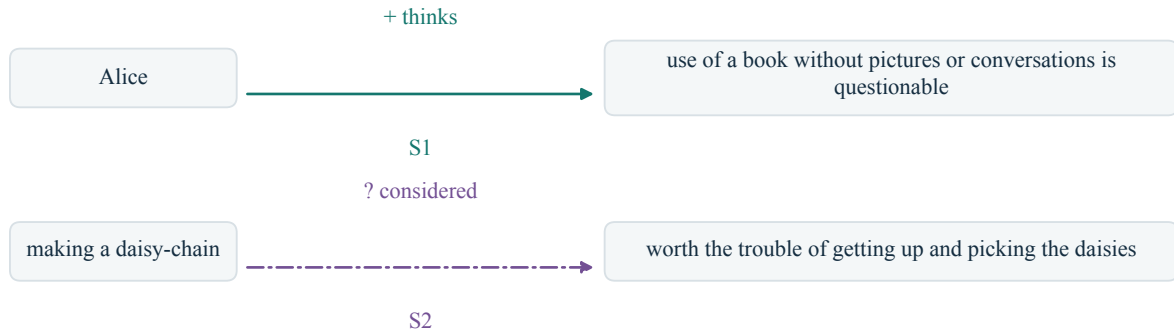
S2 ... whether the pleasure of making a daisy-chain would be worth the trouble of getting up and picking the daisies ...

B action request: sitting, reading/inspection and running



x The sister reads. Alice peeps. Same source, different participant.

B claims request: absent book content and Alice's thoughts



Final edge: holder = Alice, attitude = null. Other qualifiers null. Thought remains a sentence. Daisy-chain judgment unresolved.

+ supported meaning x unsupported ? unresolved. Source excerpted. Full passage supplied to each independent request.

Figure 3: Alice: independent contextual extraction (B) for two requests over identical complete evidence. Display: two of five action edges (two sitting edges and Rabbit running omitted) and two of three claims (book content omitted). All remain scored. Executed action question: Which sitting/location, reading/inspection and running relationships involve Alice, her sister, the book, bank and Rabbit? Exclude thoughts, feelings, appearance and book-content properties. Executed claims question: What does the passage say about the book's absent content and Alice's thoughts about the usefulness of such books and making a daisy-chain? Exclude physical actions, locations, appearance and the Rabbit; preserve questions and consideration without treating them as decisions. Source excerpts are marked. + indicates supported meaning, x an unsupported participant, and ? unresolved consideration. The thought remains the actual sentence-valued object. All bounds are null. The final record has holder Alice and null attitude.

A therefore preserves most target action meanings. Both approaches recover the later rescue, but contextual extraction omits the early capture and release. A's spare relation loses the scope of the Mouse's conditional plea, while some supported passive formulations fail strict matching (Figure 2). These networks capture useful narrative content despite incomplete agreement.

In the Alice excerpt, contextual extraction improves semantic coverage from 1 of 4 to 3 of 4, al-

though both strict F1 scores remain zero. Its combined thought statement preserves rhetorical meaning without the requested decomposition. Contextual extraction also incorrectly assigns reading to Alice instead of her sister. Whether the daisy-chain record converts considering an action's worth into judging it worthwhile remains unresolved (Figure 3). The improvement is therefore in retained meaning, with participant errors and interpretive uncertainty still visible.

Table 1: Compact-story v2 strict qualified-fact scores. A: pre-extract/select. B: contextual extraction. n includes every prediction. Targets per story: possession 5, location 4, belief/reality 4. No historical recoveries are pooled.

Story	Question	n_A	P_A	R_A	$F1_A$	n_B	P_B	R_B	$F1_B$
Harbor	Possession	5	1.000	1.000	1.000	10	0.500	1.000	0.667
Harbor	Location	4	1.000	1.000	1.000	6	0.667	1.000	0.800
Harbor	Belief/reality	5	0.000	0.000	0.000	14	0.143	0.500	0.222
Orchard	Possession	5	1.000	1.000	1.000	12	0.417	1.000	0.588
Orchard	Location	4	1.000	1.000	1.000	5	0.600	0.750	0.667
Orchard	Belief/reality	4	1.000	1.000	1.000	13	0.154	0.500	0.235
Museum	Possession	5	1.000	1.000	1.000	13	0.385	1.000	0.556
Museum	Location	4	1.000	1.000	1.000	5	0.800	1.000	0.889
Museum	Belief/reality	4	0.000	0.000	0.000	14	0.143	0.500	0.222

Table 2: Published prose: strict agreement and Codex-authored semantic assessment. A: pre-extract/select. B: contextual extraction. S/P/X/U counts supported, partial, unsupported and unresolved records. Coverage counts preserved target meanings. NA denotes parse failure, not an empty graph. All parseable predictions cite valid evidence IDs, which does not establish support.

Passage	Question	Method	n	Precision	Recall	F1	S/P/X/U	Coverage
Alice	Actions	A	2	0.500	0.200	0.286	2/0/0/0	3/5
Alice	Actions	B	5	0.400	0.400	0.400	3/1/1/0	3/5
Alice	Claims	A	4	0.000	0.000	0.000	2/2/0/0	1/4
Alice	Claims	B	3	0.000	0.000	0.000	2/0/0/1	3/4
Fable	Actions	A	8	0.625	0.625	0.625	6/1/0/1	6/8
Fable	Actions	B	5	0.400	0.250	0.308	4/1/0/0	4/8
Fable	Claims	A	1	0.000	0.000	0.000	1/0/0/0	2/5
Fable	Claims	B	4	0.000	0.000	0.000	1/3/0/0	0/5
Holmes	Actions	A	NA	NA	NA	NA	NA	NA
Holmes	Actions	B	NA	NA	NA	NA	NA	NA
Holmes	Claims	A	NA	NA	NA	NA	NA	NA
Holmes	Claims	B	3	0.000	0.000	0.000	0/1/0/2	0/5

6 DISCUSSION AND LIMITATIONS

This work advances narrative ontology development through an implemented model connecting reader intent, temporal scope and evidence inspection. It makes the construction of a narrative network an explicit part of answering a question, with relationships represented together with the information needed to interpret and verify them. Building on relation extraction (?), temporal and perspective representation (?; ?), and visual error analysis (?), the contribution is a shared framework for comparing how question availability affects extraction and selection. Relevance and interval filters provide a mechanism for reducing displayed graph size while retaining source evidence and original temporal bounds. Contextual generation explores the complementary possibility of focusing relationships during extraction itself.

The examples demonstrate useful preservation of narrative structure. Pre-extraction captures most annotated fable actions, contextual extraction increases semantic coverage in Alice, and Harbor retains ex-

plicit intervals. These findings establish concrete capabilities on which further ontology development can build. The framework also connects participant errors, lost conditions and irrelevant relationships to their source passages, making improvements in parsing and selection directly assessable. Its value lies in combining usable partial networks with an explicit account of what they preserve and omit.

7 CONCLUSION

We present a workflow connecting narrative questions, qualified relationships and evidence inspection. Comparing pre-extract/select with contextual extraction reveals useful action and thought information alongside relevance and participant errors. Selection performs better on several questions, while contextual extraction preserves more of Alice’s claims. Combining strict agreement, semantic assessment and visualization provides a concrete basis for improving narrative extraction and studying its use.

8 AI USE AND ANNOTATION PROVENANCE

OpenAI Codex (?) assisted implementation of code.
Qwen generated the simulated user prompt exchange
for query ontological searches.

APPENDIX

Qwen3-8B-AWQ revision:
4da05a8edb55c6046cce958586c33b61da07bb79.
Software: vLLM 0.10.2, Transformers 4.55.2. The
pinned template uses `enable_thinking=False`.
The concise companion indexes complete machine-
readable requests, outputs, reference alternatives and
assessments.